Pricing the URA Assistant

A feature-by-feature commercial appraisal of the *URA AI Intelligent Assistant*, conducted by **Cloake Tech Limited** under the SWEBOK Guide v4.0, Chapter 15 — *Software Engineering Economics*. Nineteen work packages measured, characterized, costed by three independent estimation techniques, and priced across Local, East African, and International market tiers.

SOVEREIGN LICENCE (LOCAL)	FIVE-YEAR COST OF OWNERSHIP	ANNUAL EQUIVALENT AT 10%	BREAK-EVEN CONVERSION
USD 480,000	USD 1,339,800	UGX 1,085 m	0.040%
UGX 1.75 bn · Tier 3, all 19 packages	UGX 4.89 bn · licence, team, infrastructure	USD 297,220 per year	of the 2.73 m dormant register/yr (at UGX 1 m/yr)

PREPARED FOR

Uganda Revenue Authority
Office of the Deputy Commissioner General

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MARKET SCHEDULE

Local (Uganda), East African (EAC), and International Enterprise & Sovereign Tiers

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Prepared for Uganda Revenue Authority – Office of the Deputy Commissioner General (and Enterprise / Sovereign Clients) **Prepared by** Cloake Tech Limited (Mpairwe Lauben · Rwemera David · Okwel Edgar Mark · Olowo Omondi Philly) **Date** 22 October 2026 · **Version** 2.0 (Commercial Edition) **Framework** SWEBOK Guide v4.0 (IEEE Computer Society, 2024) – Chapter 15 *Software Engineering Economics*, with Chapter 1 §3.2 *Economics of Quality of Service Constraints* and Chapter 7 §1.4 *Majority of Maintenance Costs*

This document supplies Cloake Tech Limited's commercial pricing schedule and SWEBOK v4.0 economic appraisal across Local (Uganda), East African (EAC), and International deployments, discharging the undertaking in §5.5 of the *URA AI Intelligent Assistant – Proposal Summary* (24 July 2026).

How to read the figures in this document

Every quantity carries a marker stating where it comes from. Nothing is presented as a fact that is actually an assumption.

MARKER	MEANING
M	**Measured.** Taken from an executed benchmark, a test run, or a count over the delivered source tree. Traceable to a named artefact in this repository.
E	**Estimated.** The team's engineering estimate, produced by a technique named in SWEBOK §15.8 and shown with its inputs so URA can re-derive or replace it.
P	**Parameter for URA.** A value only URA holds. An illustrative value is used so the arithmetic is complete; substituting URA's own figure changes the result and the method stays valid.

Currency (SWEBOK §15.9.2, multiple-currency analysis). The analysis currency is USD, with UGX shown at an assumed **UGX 3,650 / USD 1** P. This rate is not fixed and should be set at contract date. A 10% depreciation moves the recommended local Tier-3 licence fee by roughly UGX 175 million. Cloake Tech Limited recommends the contract be denominated in **one** currency, with the other shown for reference only, so exchange movement does not become a dispute.

Commercial labour rates. As a registered company, Cloake Tech Limited prices against fully-loaded corporate engineering rates (incorporating statutory taxes, NSSF, insurance, corporate overhead, and margin) across three market tiers — Local (Uganda), East Africa (EAC), and International — all marked P, and every conclusion is tested against all three (§G1, sensitivity).

PART A

Establishing the decision

A1

Purpose, and the decision this document supports

SWEBOK §15.9.1 defines a business case as "the consolidated, documented information summarizing and explaining a recommended business decision from different perspectives (cost, benefit, risk and so on) for a decision-maker." That is what this document is. It supports one decision:

> **Should URA acquire the AI Intelligent Assistant — the built, deployed and benchmarked system — together with full assignment of its intellectual property and a two-year embedded engagement of the team that built it; and if so, at what price?**

Price is the one revenue-generating element of the four Ps (SWEBOK §15.10.13); product, promotion and place are costs. This document therefore does two things a proposal cannot: it derives a **cost basis** by three independent estimation techniques, and it derives a **value basis** by the decision techniques SWEBOK prescribes for government organisations. The price sits between them, and both boundaries are shown.

§15.2.2

A2

Understanding the real problem

SWEBOK requires the real problem be established before solutions are priced, using an interrogative technique such as 5-Whys, and warns that an elicited statement is often a *solution* rather than the *problem*.

WHY	ANSWER
1. Why does URA need a taxpayer assistant?	Because the register grew from 2.62 m (FY2021/22) to 5.25 m (June 2025) while the guidance channels did not.
2. Why does that matter?	Because only 2.52 m of the 5.25 m remitted anything in FY2024/25 — 2.73 m are registered but not contributing (Auditor General, FY2024/25).
3. Why are they not contributing?	Because most non-compliance is confusion, not defiance: 76% of Ugandans report difficulty finding out what taxes they owe (Afrobarometer 886, Oct 2024).
4. Why can they not find out?	Because guidance is written, English-only and office-hours-bound, against a 74% literacy rate and 40+ living languages.
5. Why not add officers?	Because headcount scales cost linearly and reach barely at all. Reach is constrained by **language, literacy, distance and hour** — none of which an additional officer relaxes.

The real problem is a reach problem, not a staffing problem. This matters to pricing directly: it rules out "cost per officer replaced" as the valuation basis, and it makes *cost-effectiveness per taxpayer reached* the correct one (§E3).

Two things the 5-Whys exposes that a headline framing would hide:

- The binding constraint on value is **adoption**, not capacity. Measured single-GPU capacity already exceeds any plausible Year-1 demand by more than two orders of magnitude (§B5, §F5). Money spent on more capacity buys nothing; money spent on channel integration and language coverage buys everything.
- The 2.73 m dormant registrants are the population where marginal revenue per contact is highest, and they are precisely the population written English-only channels do not reach.

§15.1.8

A3

URA's business model in Drucker's terms

SWEBOK §15.1.8 adopts Drucker's four questions, and §15.2.2 asks the engineer to look closely at the client's business model to expose hidden risks and opportunities. Restated for a revenue authority, where "how do we make money" becomes "how is public value realised":

DRUCKER QUESTION	URA'S ANSWER	CONSEQUENCE FOR PRICING
Who is the customer?	Two populations, not one: 5.25 m registered taxpayers (and prospective registrants), and URA's own officers whose time the routine load consumes.	The system must be priced as serving both. The staff workbench, analytics and escalation packets are not overhead — they are half the product.
What does the customer value?	Taxpayer: a correct, checkable answer, now, in their own language. Officer: not answering the same question for the ten-thousandth time. Management: evidence of what taxpayers are actually asking.	Citations, confidence signalling and the auditable guidance record are value-bearing features, not compliance decoration.
How is value realised?	Voluntary compliance on a register where fewer than half remit; and the tax-to-GDP path from 14.2% toward 20%.	Benefit accrues per *taxpayer converted* and per *contact deflected* — both are countable, so a break-even threshold can be computed (§E2).
What is the underlying economic logic?	Guidance is a fixed-cost good: writing a correct answer once and serving it a million times costs almost nothing more than serving it once. Human guidance is a variable-cost good.	The system's economics improve superlinearly with adoption; headcount's do not. This is the crossover computed in §E3.

Hidden risk this exposes. URA's value realisation depends on *taxpayers trusting the answer*. An assistant that is fast and wrong is worth less than nothing to a revenue authority, because incorrect guidance issued in the Authority's name creates liability and erodes the voluntary-compliance base it was built to widen. This is why §B3 treats faithfulness, citation coverage and guardrail defence as **quality indicators with thresholds**, and why the evaluator–optimizer and claim-verifier packages are priced as core rather than optional.

§15.3 vs §15.4

A4

Which decision framework applies

SWEBOK is explicit: "The for-profit decision techniques don't apply when the organization's goal isn't profit — which is the case in government and nonprofit organizations" (§15.4).

URA is a government organisation. Therefore:

NOT USED	USED INSTEAD	WHERE
Minimum acceptable rate of return (§15.3.1)	— no MARR is asserted for URA	—
Internal rate of return (§15.1.5)	Benefit–cost ratio (§15.4.1)	§E2
Present-worth ranking of profit streams (§15.3)	Cost-effectiveness analysis, fixed-cost version (§15.4.2)	§E3
—	Break-even analysis (§15.5.1)	§E2, §E4
—	Multiple-attribute decision-making, additive weighting (§15.6.1)	§B4, §E1

Equivalence (§15.1.4) is still respected: all cost streams are brought to the same time frame by discounting at a **social discount rate of 10%** P and are compared over a **single planning horizon of 5 years** E (§A5). SWEBOK §15.2.5 warns specifically that using a longer time frame for one proposal than another makes the shorter one look better when it is not; all five alternatives in §E1 are therefore evaluated over the same five years.

§15.3.2, §15.3.3

A5

Planning horizon and economic life

- **Planning horizon: 5 years** E — long enough for the operate-and-maintain phase to dominate (which SWEBOK §15.10.11 says it will), short enough that estimates remain meaningful for a technology whose model tier changes annually.
- **Economic life of the GPU serving tier: 4 years** E — the point at which frozen-asset cost and rising operating cost sum to a minimum (§15.3.2). A Year-5 partial refresh line is therefore carried in the TCO (§D4).
- **Economic life of the application: not bounded by this horizon.** Because URA owns the source and weights outright, replacement (§15.3.4) can be *incremental* — a model tier, a retrieval leg or a language can be replaced without replacing the asset. SWEBOK §15.3.4 notes exactly this: "To the extent that an asset can be replaced in smaller increments, the decision-maker can consider incremental replacement options." The 49 addressable feature flags in the delivered system M are what make that possible in practice.
- **Retirement and lock-in (§15.3.5).** SWEBOK notes retirement decisions "can be influenced by lock-in factors such as technology dependency and high exit costs." Under the proposed IP assignment, URA's exit cost is **zero**: no licence to terminate, no weights to hand back, no vendor whose withdrawal removes the service. This is a priced-in property of Alternative A4 and a decisive difference from A2 (§E1).

PART B

Analysis and evaluation of every feature

B1

Method

The system is decomposed into **19 work packages** covering the entire delivered surface. Each package is evaluated on five axes, and each axis is a named SWEBOK technique:

1. **Evidence** – what exists, counted over the source tree `M`.
2. **Replacement effort** – engineer-months to rebuild, by decomposition (`§15.8.3`) `E`.
3. **Quality indicator** – a measured indicator, normalised (`§15.7.4`) `M`.
4. **Impact indicator** – contribution to a named URA business goal (`§15.7.4`) `E`.
5. **Characterization** – the SIPAC state per Figure 15.5 (`§15.7.5`).

Normalisation follows `§15.7.5`: indicators are standardised to **[-1, +1]** against a stated threshold, where **0 means on target**, **+1 means 100% over target**, and **-1 means 100% under target**. For a higher-is-better indicator with measurement M and threshold T, `norm = (M - T)/T`; for lower-is-better, `norm = (T - M)/T`; both clipped to [-1, +1].

B2

Delivered system — counted scale

All counts are over the delivered source tree at commit `5c1c7cd` (21 Aug 2026) `M`.

DIMENSION	COUNT
Backend application code (`App/backend/app`, Python)	58,693 lines
Frontend application code (`App/frontend`, TypeScript/TSX)	21,189 lines
ML / MLOps pipelines (`ml/`)	25,023 lines
Operational scripts (`scripts/`)	9,546 lines
Automated test code (backend, integration, chaos, frontend, E2E)	≈ 37,100 lines
Total first-party code	**≈ 149,900 lines**
Documentation (58 documents)	16,765 lines
CI/CD pipeline definitions (21 workflows)	6,975 lines
HTTP and WebSocket routes	73
Registered agent tools (of which 8 statutory calculators)	25
Addressable feature flags	49
Automated tests (backend functions / frontend units / E2E suites)	2,344 / 165 / 14
Indexed knowledge passages	5,071
Source corpus	533 MB across 1,334 files, incl. 148 PDFs
Delivery record	899 commits, 255 merged pull requests, 24 Dec 2025 – 21 Aug 2026

B3

The 19 work packages — effort, quality, impact, state

Effort is **engineer-months (EM)** to rebuild the package to the delivered standard – tested, documented and compliance-mapped – by a competent professional team `E`. It is *not* the team's own expenditure, which is a sunk cost and is treated separately in `§D3`.

Quality indicators are drawn from the executed benchmark reports of 21 Aug 2026 in `docs/Reports/` `M`. Impact indicators are the team's assessment against the five URA business goals of `§C1` `E`.

#	WORK PACKAGE	PRINCIPAL EVIDENCE	EM	QUALITY INDICATOR (MEASURED)	QVAL	IVAL	KAVAL	STATE
1	**Retrieval & RAG core** – hybrid dense (BGE-M3) + BM25 + RRF, cross-encoder rerank, semantic cache, query rewrite, query decomposition, corrective re-retrieval, HyDE, circuit breaker	9,300 LOC; `retriever.py`, `service.py`, `cache.py`, `query.py`, `corrective_rag.py`	14	Faithfulness 0.93 vs 0.70 target; citation coverage 10/10	+0.33	+0.60	**+0.47**	Stable
2	**Generation & model serving** – Sunflower-14B-FP8, vLLM, 4-bit BnB, LoRA routing, quantised variants, speculative decoding, prefix caching, per-turn model tiering	3,100 LOC; `llm.py`, `providers/`	10	p50 2,240 ms full-stack EN; 0% error to c=1,000	+0.55	+0.55	**+0.55**	Stable
3	**Agentic orchestration** – supervisor/specialist graph, 7 routes, LangGraph orchestrator, loop control, evaluator–optimizer, LLM tiebreak	5,800 LOC; `agents/`, `graph/build.py`	11	Intent determination 100% across en/lg/sw	+0.11	+0.60	**+0.36**	Stable
4	**Tool & MCP layer** – 25 registered tools incl. 8 statutory calculators (PAYE, VAT, WHT, customs, rental, CGT, corporation tax, VAT-registration test), MCP client/transport/policy, Tool-RAG selection, risk tiers and scope declarations	7,260 LOC; `tools/`, `mcp/`, `calculator_router.py`, `tax/tables.py`	12	Arithmetic precision 100%; schema conformance 100%	+0.11	+0.80	**+0.46**	Stable
5	**Statutory knowledge graph** – effective-dated rate graph, rate history, graph-fusion retrieval leg, shadow scoring	1,493 LOC; `graph/`	4	Shadow-scored; not yet in the answer path	–	+0.40	**+0.40**	Acceptable Impact (Case 3)
6	**Multilingual & translation** – translate-and-retrieve, Sunbird MT, locale-specific supervisor routing, Ugandan accent detection (5 profiles)	1,500 LOC; `sunbird.py`, `accent_detector.py`	6	FAQ accuracy 100% en/lg/sw; lg p50 619 ms	+0.11	+1.00	**+0.56**	Stable
7	**Voice pipeline** – Whisper ASR, Piper/Spark TTS, streaming WebSocket transport, VAD with hysteresis, barge-in, native voice-to-voice, CosyVoice2 token-level TTS, speculative prefetch, voice consent	6,600 LOC; `speech_service.py`, `voice_stream*.py`, `voice_ws*.py`, `native_voice/`	16	Transcript accuracy 100% en/lg/sw; TTS p50 239–415 ms	+0.15	+1.00	**+0.58**	Stable
8	**Vision & document intelligence** – OCR, document classifier, table structuring, glyph fusion, 5 formats to 40 MiB, malware scan, PDF guards, session-bound passage injection	4,800 LOC; `vision/`, `documents.py`, `ocr_service.py`	11	40 MiB at 4.82 MB/s; 100% TIN/UGX/date extraction	+0.61	+0.60	**+0.61**	Stable
9	**Safety & guardrails** – OWASP LLM Top 10 input/output guards, PII redaction (TIN/NIN/card/phone), prompt-leak detection, XSS sanitisation, retrieved-passage scrubbing, claim verifier, entailment, authority check	2,000 LOC; `guardrails.py`, `claim_verifier.py`, `entailment.py`, `authority.py`	8	Defence rate 100% vs 95% target, retained at c=250 and over 1,500-probe soak	+0.05	+1.00	**+0.53**	Stable
10	**Identity, consent & data-subject rights** – OIDC/JWT, role claims, RLS multi-tenancy, UDPA 2019 export and erasure, consent grant/withdraw ledger	2,100 LOC; `auth/`, `tenancy.py`, `voice_consent.py`, `retention.py`	7	0 cross-tenant violations in 600 mixed concurrent ops; UDPA export 6.5 ms	+1.00	+0.80	**+0.90**	Stable
11	**Audit & compliance** – hash-chained audit ledger with Merkle verifier, AI risk manifest, compliance gate, production-readiness gate	1,300 LOC; `audit/`, `governance/`	6	100% gate verification across NIST AI RMF, ISO/IEC 42001, EU AI Act, OWASP LLM 2026; 28/28 STRIDE threats mitigated	+0.05	+0.80	**+0.43**	Stable
12	**Human-in-the-loop & staff workbench** – HITL routing, SLA tracking, ticket WebSocket with presence, structured handoff packets, notifications, CMS answer overrides, 5 staff pages	3,500 LOC; `hitl_routing.py`, `ticket_ws.py`, `cms.py`, `/admin/*`	9	Staff admin p50 24.96 ms under mixed load; escalation packets carry full context	+0.50	+0.80	**+0.65**	Stable
13	**Analytics, observability & evaluation** – OpenTelemetry GenAI per-stage spans, Prometheus metrics, dashboards, evaluation harness with 9 quality gates, corpus freshness, index lifecycle	2,600 LOC; `tracing.py`, `analytics.py`, `evaluation.py`, `freshness.py`	8	42/42 CI pipelines green (100% pass rate); 9/9 evaluation quality gates passed	+0.11	+1.00	**+0.56**	Stable

#	WORK PACKAGE	PRINCIPAL EVIDENCE	EM	QUALITY INDICATOR (MEASURED)	QVAL	IVAL	KAVAL	STATE
14	**Offline / edge / mobile** – FAISS + ONNX offline RAG, hash-diff delta sync, SHA-256-verified signed bundles, on-device vector search	1,300 LOC; `offline_rag.py`, `offline_sync.py`, `offline_bundle.py`	6	Bundle target ≤ 800 MB enforced in CI; on-device p95 target 180 ms not yet field-measured	-0.10	+0.60	**+0.25**	Evolving
15	**Content operations & corpus engineering** – crawler, PDF and FAQ ingestion, publications pipeline, topic taxonomy, Qdrant backup/restore	2,200 LOC + 533 MB corpus; `crawl_corpus.py`, `pdf_corpus.py`, `faq_corpus.py`	9	5,071 passages indexed from 148 PDFs; freshness check automated	+0.21	+0.80	**+0.51**	Stable
16	**Export & reporting** – branded conversation PDF, tax-summary PDF, XLSX/DOCX/CSV exports, UDPA portability JSON	1,500 LOC; `artifact_export.py`, `pdf_export.py`	4	Exports 6.5–337.5 ms, sub-second across 4 formats	+0.66	+0.40	**+0.53**	Stable
17	**Personalisation & memory** – working / episodic / semantic memory, decay, fact extraction, reminders	1,300 LOC; `memory/`, `reminders.py`	5	Consent-gated; behind `memory_enabled`, not yet measured in production	–	+0.40	**+0.40**	Acceptable Impact (Case 3)
18	**Taxpayer web experience & accessibility** – 12 pages, 45 components, WCAG 2.2 AA, responsive, voice-first mode, streaming reply surface, citation and confidence display	≈ 17,000 LOC; `App/frontend/src`	14	14 Playwright E2E suites green; WCAG 2.2 AA gate green in CI	+0.17	+1.00	**+0.59**	Stable
19	**Platform, MLOps & assurance** – Docker GPU image, compose overlays, 21 CI/CD pipelines, SBOM (CycloneDX 1.6), SLSA provenance, Kaggle training pipeline, deployment automation, 58 documents, full standards mapping	6,975 lines CI + 16,765 lines docs + 9,546 lines scripts	18	12/12 standards mapped with gap status closed; 0 exposed credentials (Gitleaks, TruffleHog)	+0.11	+0.60	**+0.36**	Stable
	Total		**178**					

Reading the characterization (SWEBOK §15.7.5, Figure 15.5). Thresholds are set at 0 for both axes – an indicator at target is acceptable, below target is not. KAval is computed by the three existential rules of §15.7.5: where both a quality and an impact indicator exist, KAval = (Qval + Ival)/2; where only one exists, KAval is that one and the asset falls into Case 2 or Case 3 rather than a quadrant.

The thresholds, stated in full. So that every Qval above can be recomputed independently rather than taken on trust, the threshold *T* behind each one is set out here. Measurements are M, drawn from the benchmark reports of 21 Aug 2026; thresholds are E, set by the team before the measurements were taken and open to renegotiation with URA.

#	PRIMARY QUALITY INDICATOR, AS MEASURED M	THRESHOLD T E	DIRECTION	QVAL
1	Faithfulness 0.93	0.70 – the evaluation harness's own release gate	higher-better	+0.33
2	Full-stack p50 2,240 ms	5,000 ms – the conversational-turn budget of §B5	lower-better	+0.55
3	Intent-determination accuracy 100%	90%	higher-better	+0.11
4	Arithmetic precision and schema conformance 100%	90%	higher-better	+0.11
5	*none in the answer path*	–	–	*Case 3*
6	Multilingual FAQ accuracy 100%	90%	higher-better	+0.11
7	Transcript accuracy 100%	87%, i.e. WER ≤ 13%	higher-better	+0.15
8	Document ingest 4.82 MB/s	3.0 MB/s – 40 MiB inside a 14 s upload budget	higher-better	+0.61
9	Guardrail defence rate 100%	95% – safety-critical, raised above the 90% default	higher-better	+0.05
10	0 cross-tenant violations in 600 mixed concurrent operations	0 – zero-defect	at ceiling	+1.00
11	Compliance-gate verification 100%	95% – safety-critical	higher-better	+0.05
12	Staff-workbench p50 24.96 ms	50 ms – interactive-UI budget	lower-better	+0.50
13	CI pipeline pass rate 100% (42 of 42)	90%	higher-better	+0.11
14	On-device p95 **not field-measured**	180 ms, CI-enforced only	*unverified*	–0.10
15	5,071 passages indexed	4,200 – corpus-coverage target	higher-better	+0.21
16	Slowest export 337.5 ms	1,000 ms – sub-second target across all four formats	lower-better	+0.66
17	*none in the answer path*	–	–	*Case 3*
18	14 Playwright E2E suites green	12 – one per taxpayer-facing page group	higher-better	+0.17
19	Standards mapped with gap status closed, 12 of 12 = 100%	90%	higher-better	+0.11

Four scoring conventions govern that table, and they are applied without exception:

1. **Rate indicators** are scored against a pass-rate threshold. The default is 90%; it is raised to 95% for the two safety-critical packages (9 and 11), where a one-in-twenty failure is not an acceptable operating point.
2. **Zero-defect indicators measured at their ceiling** score **+1.00**. A count of zero cannot be exceeded, so no proportional scale applies to it.
3. **A target enforced in CI but not verified in the field** scores **–0.10**. This is a deliberate penalty rather than a measurement, and it is the reason package 14 is characterized *Evolving* instead of *Stable*.
4. **One primary indicator per package.** Corroborating results – package 19's 0 exposed credentials and 28/28 STRIDE threats mitigated, package 13's 9/9 evaluation gates, package 1's 10/10 citation coverage – are recorded as evidence but are **not** double-counted into Qval. Where a package has several defensible indicators, the **least flattering** one is taken as primary. This makes the quality scores conservative by construction, which is the appropriate bias in a document that is also asking URA for money.

All values are carried to two decimal places and rounded half-up. On that convention every Qval in the table above and every KAval in §B3 reproduces exactly from the measurement, the threshold and the two

normalisation formulae of §B1 – there is no adjusted or judgement-weighted figure anywhere in the scoring except the single stated –0.10 convention for package 14.

Two results are worth URA's attention:

- **Package 14 (offline / edge) is the only package characterized "Evolving"** – high impact, quality below threshold, because its on-device latency target has been enforced in CI but not field-measured on real mid-range Ugandan Android hardware. It is the one package where the team recommends URA hold payment against a measured field gate (§F3, milestone M2).
- **Packages 5 and 17 sit in Case 3** – impact indicators only. Both are behind default-off flags. They are delivered and tested but have not yet earned a quality measurement in the answer path, and they are priced as Tier-3 scope accordingly (§F6).

§15.6.1, §15.7.7

B4

Multiple-attribute ranking of the packages

SWEBOK §15.7.7 lists the criteria by which software products supporting intangible assets should be prioritised: impact on business goals, characterization reached, impact on competitors, impact on the business model, cost to implement, time to implement, and complexity. Those seven criteria are used exactly, by additive weighting (a compensatory technique, §15.6.1), scored 0–5.

Weights E: goal impact 0.30, characterization 0.15, competitive position 0.10, business-model impact 0.15, cost 0.10, time 0.10, complexity 0.10. Cost, time and complexity are scored inversely (cheap/fast/simple scores high).

RANK	PACKAGE	SCORE	WHY IT RANKS HERE
1	9 · Safety & guardrails	4.35	Highest goal impact of any package: an assistant that can be talked into wrong guidance is a liability, not an asset. Cheap relative to consequence.
2	1 · Retrieval & RAG core	4.25	The grounded, cited answer *is* the product. Everything else is delivery.
3	7 · Voice pipeline	4.20	The only package that reaches the ~26% of adults for whom reading is a barrier. Also the sole regional differentiator (§E1).
4	6 · Multilingual & translation	4.15	Same reach argument, at lower cost and complexity than voice.
5	10 · Identity, consent & rights	4.05	UDPA 2019 is not optional; measured 0 cross-tenant violations makes it the highest-Qval package.
6	12 · HITL & staff workbench	3.95	Where officer-time benefit is actually realised, and where URA's own staff touch the system.
7	4 · Tool & MCP layer	3.90	Turns "here is the rule" into "here is your figure". High impact, moderate cost.
8	18 · Taxpayer web experience	3.80	Adoption is the binding constraint; this is the adoption surface.
9	13 · Analytics & evaluation	3.75	The management-intelligence benefit URA cannot get from existing channels.
10	2 · Generation & model serving	3.70	Essential but substitutable – the model tier can be swapped without touching the rest.
11	19 · Platform, MLOps & assurance	3.55	High cost, invisible to taxpayers, indispensable to operating the thing.
12	11 · Audit & compliance	3.50	Enables the auditable-guidance-record benefit; modest cost.
13	3 · Agentic orchestration	3.40	Raises ceiling quality; the system answers without it.
14	15 · Content operations	3.35	Ongoing rather than one-off; belongs with URA content staff after handover.
15	8 · Vision & document intelligence	3.10	Strong capability, but its highest-value use is the *internal* work-product extension, not taxpayer Q&A.
16	16 · Export & reporting	2.95	Cheap, useful, not load-bearing.
17	17 · Personalisation & memory	2.60	Case-3 asset. Real value, unproven in production, consent-heavy.

RANK	PACKAGE	SCORE	WHY IT RANKS HERE
18	5 · Statutory knowledge graph	2.55	Case-3 asset. The right long-term answer to effective-dated rates; shadow-mode today.
19	14 · Offline / edge / mobile	2.40	Highest reach potential per shilling in rural districts, but the only "Evolving" package.

Reproducing the ranking. Each package is scored 0–5 against each of the seven criteria and the scores combined by the weights above. The full 19 × 7 matrix is a working sheet rather than a result, and it can be reconstructed by any reader from three things already in this document: the weights stated above, the **EM** column of §B3 (which drives the cost, time and complexity criteria), and the **State** column of §B3 (which drives the characterization criterion). Note that the characterization criterion is scored on the *state reached* — Stable, Evolving, Case 2, Case 3, Warning — and not on the KAval magnitude, so the ranking is stable under small revisions to individual indicators. Differences inside 0.05 should be read as ties: the ranking is intended to separate tiers of importance, not to order packages 6 and 7 against one another.

This ranking is what makes a **scope-tiered price** possible (§F6) rather than a single take-it-or-leave-it figure. It is also the honest answer to "what should URA buy first if the budget is constrained": packages 9, 1, 7, 6, 10.

§1.3.2

B5

The economics of the quality-of-service constraints

SWEBOK Chapter 1 §3.2 warns that quality-of-service constraints are challenging "because engineers do not consider them from an economic perspective," and gives three points to locate: the **perfection point** (beyond which extra performance carries no benefit), the **fail point** (below which there is no further loss because the user has already left), and the **most cost-effective performance level** (maximum positive difference between value and cost).

Located against the measured single-GPU concurrency curve M (docs/Reports/SINGLE_GPU_CAPACITY_LIMITS_REPORT_2026-08-21.md):

CONCURRENCY	P50	P95	ERRORS	ECONOMIC READING
c = 10	30.4 ms	64.4 ms	0.0%	**Past the perfection point.** A taxpayer cannot read faster than they can read; 30 ms and 400 ms are indistinguishable in use. Capacity spent here is wasted.
c = 100	415.3 ms	461.3 ms	0.0%	**Most cost-effective performance level.** Sub-500 ms, 214 RPS sustained, zero errors.
c = 250	1,498.7 ms	1,862.5 ms	0.0%	Acceptable degradation. Value falling, cost flat.
c = 500	2,861.2 ms	3,528.3 ms	0.0%	Approaching abandonment.
c = 1,000	4,657.0 ms	5,985.1 ms	0.0%	**At the fail point.** p95 ≈ 6 s is where a taxpayer stops waiting; beyond it there is no further benefit to lose.

Perfection point ≈ 500 ms p50 E. Fail point ≈ 6 s p95 [M, at the measured ceiling]. Economically optimal operating point: c ≤ 100 per GPU.

This single result does real pricing work. It fixes the GPU count from the demand model instead of from procurement instinct (§F5), and it says plainly that URA should **not** pay for additional serving capacity to chase latency below 500 ms — that expenditure buys performance past the point at which any taxpayer can use it.

PART C

Intangible assets

§15.7.1–7.3

C1

URA's business goals and the intangible assets behind them

SWEBOK §15.7.1 gives five generic business-goal categories. Instantiated for URA:

SWEBOK GOAL CATEGORY	URA INSTANTIATION	IMPORTANCE (1–5, §15.7.2)
Growth and continuity	Convert the 2.73 m dormant registrants into remitting taxpayers	**5**
Meeting financial objectives	Tax-to-GDP from 14.2% toward 20% (Minister of Finance, 2 Jul 2026)	**5**
Responsibility to employees	Relieve frontline and contact-centre officers of repetitive load; reduce burnout	**4**
Responsibility to society	Equitable access to tax information across language, literacy and distance	**5**
Managing market position	Regional leadership in digital tax administration	**3**

§15.7.2 asks for the intangible assets that serve as levers on those goals, and §15.7.3 for the software products that support them. The mapping – and this is the core of the value argument – is that **URA already owns the intangible assets; what it lacks is the delivery mechanism.**

INTANGIBLE ASSET URA ALREADY HOLDS	ITS CURRENT CONDITION	SOFTWARE PACKAGES THAT ACTIVATE IT
Published guidance corpus (FAQs, publications, handbooks, 148 PDFs indexed)	Written, English, static, requires the taxpayer to find and read it	1, 15, 18
Officer tacit knowledge of procedure	Non-transferable, unrecorded, lost when the call ends	4, 12, 15
Statutory rate and threshold knowledge	Correct, but effective-dating lives in officers' heads	4, 5
Escalation and triage practice	Works, but every escalation restarts from zero	12
Taxpayer-education material	Reaches taxpayers when URA can deliver it, not when the question arises	6, 7, 18
Service-channel playbooks	Cannot hold a back-and-forth about one taxpayer's specific question	3, 12
Record of guidance issued in URA's name	**Does not exist.** Counter and phone guidance leaves no trace	11, 13

Two observations that bear directly on price:

1. **Most of the value is unlocking assets URA has already paid for.** The guidance corpus, the procedural knowledge and the rate tables are existing investments whose return is limited by a delivery channel, not by their content. That is why benefit accrues at a scale far above the cost of the delivery mechanism (§E2).
2. **One asset does not exist yet and is created outright.** The auditable record of guidance issued in the Authority's name (packages 11, 13) is a new intangible asset, not an improvement to an old one. SWEBOK §15.1.7 admits precisely this class of value – "grants rights and economic benefits to its owner" without physical substance.

§15.7.6

C2

Linking the assets to the business model

§15.7.6 asks that the client's business model be visualised enriched with the intangible assets' status, so leadership can see which proposed solution generates the most value. Stated as a table, highest KAval first:

KAVAL	PACKAGES	ASSET STATE	WHAT IT MEANS FOR URA
+0.90	10	Stable	Statutory obligations (UDPA 2019) are met with measured evidence, not assertion
+0.65	12	Stable	Officer-time benefit is realisable on day one
+0.59 – 0.61	8, 18	Stable	Document intelligence and the adoption surface are both above target
+0.51 – 0.58	2, 6, 7, 9, 13, 15, 16	Stable	The reach and safety core is above target across the board
+0.36 – 0.47	1, 3, 4, 11, 19	Stable	Above target, with the most headroom for improvement against URA's own corpus
+0.40	5, 17	Case 3	Impact established, quality not yet earned in the answer path
+0.25	14	**Evolving**	Impact high, quality below threshold — the one field-measurement gate

No package is characterized "Warning" (low impact *and* low quality). Every package either serves a goal with measured quality above threshold, or is explicitly flagged as unproven and priced as such.

PART D

Cost

§15.8

D1

Three independent estimates

SWEBOK §15.8.5 is unambiguous: "when the consequences of a wrong decision are significant, investing extra effort in developing more than one estimate can be worthwhile… Convergence suggests the individual estimates are probably accurate." A national-scale system for a revenue authority is such a case. Three techniques were used.

Estimate 1 — Decomposition

Bottom-up from the 19 work packages of §B3. The cross-cutting factors §15.8.3 warns about — requirements work, integration, testing, user documentation — are carried inside packages 13 and 19 rather than omitted.

178 EM, range **140–220 EM** E. Implied productivity: 149,900 LOC ÷ 178 EM = **842 lines per engineer-month** of tested, documented, standards-mapped code.

Estimate 2 — Parametric, COCOMO II post-architecture

§15.8.4 notes parametric estimates are "typically the most accurate, the most defendable and the easiest to use, provided the equation has been developed and validated."

- Delivered production source: 116,600 SLOC (excluding test code) M
- Reuse adjustment: 60% new at weight 1.0, 40% integration-over-mature-OSS at AAF 0.35 → **86.3 KESLOC** equivalent E
- Scale factors: PREC low (3.72), FLEX nominal (3.04), RESL high (2.83), TEAM very high (1.10), PMAT nominal (4.68) → ΣSF = 15.37 → **E = 0.91 + 0.1537 = 1.0637**
- Nominal effort: 2.94 × 86.3^1.0637 = **336.9 PM**
- Effort multipliers: RELY 1.26 · DATA 1.14 · CPLX 1.34 · RUSE 1.07 · DOCU 1.11 · PVOL 1.15 · ACAP 0.71 · PCAP 0.76 · PCON 0.90 · PLEX 0.91 · LTEX 0.91 · TOOL 0.78 · SITE 0.80 → **∏EM = 0.660**

222 EM E.

Estimate 3 — Analogy on delivered increments

§15.8.2 requires "a suitable analogy for which actual results are known." The known actual is this project's own delivery record: **255 merged pull requests** M, at the phase-scale granularity visible in the history (each carrying design, implementation, tests, docs and CI). At a professional rate of **0.5–0.9 EM per merged increment** E, central 0.7:

179 EM, range **128–230 EM** E.

Stated limitation: this is the weakest of the three legs. SWEBOK §15.8.2 requires an analogy whose actual results are known, and no validated public cost figure exists for a comparable revenue-authority conversational-AI programme — Kenya's *Shuru* (launched 2 Apr 2026) has no published cost. The leg is therefore an internal analogy, and it is reported for convergence only.

Convergence

TECHNIQUE	SWEBOK §	ESTIMATE	RANGE
Decomposition	15.8.3	178 EM	140–220
Parametric (COCOMO II)	15.8.4	222 EM	–
Analogy on delivered increments	15.8.2	179 EM	128–230
Converged (mean)	**15.8.5**	**193 EM**	**140–230**

The three estimates span 178–222 EM – a half-span of ±22 EM, or **±11% of the converged mean of 193**. Under §15.8.5 this is convergence, and the converged figure is used. **Replacement effort: 193 engineer-months.**

D2

Commercial replacement cost at market rates

Converged effort valued at three corporate market rate scenarios, all P:

RATE SCENARIO	USD / EM	193 EM (USD)	193 EM (UGX)
Local (Ugandan corporate engineering rate)	3,500	675,500	2.47 bn
East African (EAC regional specialist vendor)	7,000	1,351,000	4.93 bn
International (Global AI-systems integrator)	15,000	2,895,000	10.57 bn

These are **reference ceilings**, not the price. They answer one question: what it would cost URA (or an equivalent regional or international authority) to obtain this capability by any route other than licensing from Cloake Tech Limited.

§15.10.2

D3

Sunk cost is deliberately excluded

SWEBOK §15.10.2 states plainly: *"Sunk cost refers to unrecoverable expenses that have occurred, which can cause emotional hurdles looking forward. From a traditional economics viewpoint, sunk costs should not be considered in decision-making."*

Cloake Tech Limited's prior R&D expenditure – four principal engineers across eight months and 899+ commits of internal product engineering – is a sunk cost. **It is not the basis of the fee, and the client does not price against historical R&D.** It is recorded here once, for completeness, and then set aside.

The economically correct basis, per §15.10.2, is **opportunity cost**: the cost of the alternative URA must forgo. That is the replacement cost of §D2 and the alternatives of §E1 – and it is what the fee is set against.

§15.10.2, §15.10.11

D4

Total cost of ownership over the planning horizon

SWEBOK §15.10.2 defines TCO as "the total cost for acquiring that product, activating it and keeping it running," and notes it "holds true especially for software because there are many not-so-obvious costs related to SPLC activities after initial product development." §15.10.11 adds that the operate-maintain-retire activities "consume more total effort and other resources than the SDLC activities," and Chapter 7 §1.4 that **over 80% of maintenance is enhancement and adaptation, not fault-fixing**. All three are honoured below.

All figures USD E, with P corporate rate assumptions as stated (Local Uganda Sovereign Reference Case).

COST LINE	Y1	Y2	Y3	Y4	Y5	TOTAL

COST LINE	Y1	Y2	Y3	Y4	Y5	TOTAL
Sovereign licence & IP grant (milestoned, §F3)	336,000	144,000	–	–	–	**480,000**
Embedded corporate engagement, 4 specialists (§F4)	216,000	216,000	–	–	–	**432,000**
Infrastructure capex – 4-GPU production + 2-GPU staging; Y5 partial refresh	82,000	–	–	–	20,000	**102,000**
Infrastructure operating – power, cooling, hosting, network, ops share	14,000	14,000	14,000	14,000	14,000	**70,000**
Post-handover maintenance – 2 URA FTE from Y3	–	–	33,600	33,600	33,600	**100,800**
Content operations – corpus freshness, 1 URA FTE	12,000	12,000	12,000	12,000	12,000	**60,000**
Model refresh and re-evaluation, annual	8,000	8,000	8,000	8,000	8,000	**40,000**
Independent assurance – penetration test, evaluation audit	15,000	10,000	10,000	10,000	10,000	**55,000**
Annual total	**683,000**	**404,000**	**77,600**	**77,600**	**97,600**	**1,339,800**

Five-year TCO: USD 1,339,800 ≈ UGX 4.89 bn E**.**

Applying equivalence (§15.1.4) at a 10% social discount rate P:

- **Present worth of cost: USD 1,126,699 ≈ UGX 4.11 bn**
- **Annual equivalent cost: USD 297,220/yr ≈ UGX 1,085 m/yr** (capital recovery factor 0.263797)

The annual-equivalent figure of **UGX 1,085 m/yr** is used for every benefit comparison in Part E, because SWEBOK §15.1.4 permits comparison only between cash flows expressed in the same time frame.

The SPLC warning, quantified. Post-licence lines total USD 427,800 in direct operations plus USD 432,000 in embedded engineering – **64.2% of the five-year TCO**, and a rising share every year thereafter. This is exactly what §15.10.11 predicts, and it is the single most common omission in software procurement. URA should budget for it explicitly. Per Chapter 7 §1.4, the great majority of that spend will be **enhancement and adaptation** – new tax measures, new languages, new channels – not defect repair, and it should be planned as capability development rather than as a warranty reserve.

§15.4, §15.8, §15.10.13

D5

Government rate benchmarks and FY 2026/27 fiscal parameters for determining system price

For Government of Uganda (GoU), PPDA, NITA-U, and MoFPED (**Vote 141**) procurement appraisal, the price of a sovereign enterprise system is determined by cross-checking three independent baselines: (1) **URA's FY 2026/27 macro-fiscal mandate**, (2) **official public-sector and corporate engineering rate schedules**, and (3) **subsystem replacement cost anchored in FY 2026/27 statutory valuation complexity**.

1. URA FY 2026/27 macro-fiscal baseline (Vote 141)

STRATEGIC INDICATOR	VERIFIED FY 2026/27 FIGURE	SYSTEM VALUATION & PRICING LEVER
FY 2026/27 National Budget	**UGX 84.392 Trillion** M	Establishes sovereign public-finance scale.
URA Domestic Revenue Target (Vote 141)	**UGX 40.09 Trillion** (≈ USD 10.84 bn at UGX 3,700/USD) M	Every **0.05%** lift in voluntary on-time filing and payment yields **+UGX 20.0 bn/yr** (≈ USD 5.41 m/yr).
Registered Taxpayer Base & Inactive Gap	**5.7 m taxpayers** (3.3 m active; **2.1 m / 37% inactive**) M	Reactivating **1.5%** of inactive presumptive taxpayers (31,500 at UGX 200,000 Second Schedule tax) unlocks **+UGX 6.3 bn/yr**; 5-locale (en, lg, sw, nyn, ach) voice/chat + Portal Co-Pilot yields **UGX 12.0 bn – 32.0 bn/yr** in compliance lift.

STRATEGIC INDICATOR	VERIFIED FY 2026/27 FIGURE	SYSTEM VALUATION & PRICING LEVER
CXP & MZO Officer Productivity	**3.6 m – 4.8 m contacts/yr** across `0800 117 000`, web, email, MZO `E`	**2.5 m autonomous resolutions/yr** × **UGX 4,500** net officer cost saved = **UGX 11.25 bn/yr** (≈ USD 3.04 m/yr) in released officer capacity, for a combined annual economic value of **UGX 23.25 bn – 43.25 bn/yr** (USD 6.28 m – 11.68 m/yr).
Vote 141 Operational Budget	**UGX 917.3 Billion** (≈ USD 247.9 m) `M`	Turnkey Year-1 system price (USD 1.17 m / UGX 4.33 bn including integration) is **0.47%** of Vote 141 (`0.011%` of the revenue target) with a < **2.5-month payback**.

2. Government and market rate benchmarks for determining system price

BENCHMARK CATEGORY	RATE / PARAMETER	HOW IT DETERMINES SYSTEM PRICE
GoU / URA Internal Officer Rate (`U4–U1E` Equivalence)	**UGX 3.33 m / month** (**UGX 40 m / year** ≈ USD 10,810/yr; **UGX 4,500 – 5,333 / contact**) `P`	Sets the public-sector labour alternative (A1) in `§E3`. Above **557 contacts/day**, the AI assistant is strictly more cost-effective per taxpayer served than adding human headcount.
PPDA Local Corporate Engineering Rate (Uganda)	**USD 3,500 / EM** (UGX 12.78 m – 12.95 m / EM, fully loaded with 30% CIT, 10% employer NSSF, 6% WHT, overhead) `P`	Multiplied by the **193 EM** converged effort (`§D1`) to establish the **USD 675,500** local standalone manufacturing floor. The **USD 480,000** Tier-3 base sovereign licence is set at **71%** of this floor (**USD 2,487 / EM**) because Cloake Tech retains non-competing regional IP.
NITA-U / PPDA Embedded Specialist Rate Card	**USD 3,200 / USD 4,500 / USD 6,500 per person-month** (Associate / Mid / Senior AI Architect) `P`	Determines the 24-month capability-transfer price (`§F4`): **4 Mid Specialists** × **24 months** × **USD 4,500** = **USD 432,000** (UGX 1.58 bn), transitioning operations to URA FTEs by Year 3.
EAC Regional Specialist Vendor Rate	**USD 7,000 / EM** (≈ UGX 25.6 m – 25.9 m / EM) `P`	Establishes the **USD 1,351,000** (≈ UGX 4.93 bn – 5.00 bn) regional replacement ceiling, aligning with the 4-subsystem appraisal of **USD 1,400,000** (UGX 5.18 bn).
International Tier-1 Systems Integrator Rate	**USD 15,000 / EM** (≈ UGX 54.8 m – 55.5 m / EM) `P`	Establishes the **USD 2,895,000** (UGX 10.57 bn) global integrator ceiling; the local sovereign licence fee represents an **83.4% discount** below international procurement.
NITA-U Agency Vote & Per-Citizen Benchmarks	**< 1.0% of Agency Vote · USD 0.11 (UGX 422) / TIN / yr** `M`	Keeps capital outlay within NITA-U ICT vote-share ceilings (`0.47%` of Vote 141 for turnkey; `0.26%` for Managed SaaS at USD 650,000/yr).
Government AMC, Social Discount & Statutory Tax Rates (`FY2026-27.json`)	**20% p.a. AMC · 10% MoFPED Social Discount Rate · 18% VAT · 6% Resident WHT vs. 15% Non-Res. WHT + 5% DST** `M`	Annual statutory maintenance is set at **20% AMC** (within PPDA's 15%–22% software support norm). Contracting a resident Ugandan company retains **18% VAT** and **6% WHT** (*ITA Cap 338 s.119*) domestically, avoiding the **15% non-resident WHT** (*s.85*) and **5% Digital Services Tax** (*s.86A*) gross-up charged by foreign cloud vendors.

3. Component-level replacement valuation and embedded FY 2026/27 statutory valuation rules (`FY2026-27.json`)

DELIVERED SUBSYSTEM	TECHNICAL SCOPE & EMBEDDED FY 2026/27 STATUTORY VALUATION RULES	REPLACEMENT VALUE (USD)	REPLACEMENT VALUE (UGX @ 3,700)

DELIVERED SUBSYSTEM	TECHNICAL SCOPE & EMBEDDED FY 2026/27 STATUTORY VALUATION RULES	REPLACEMENT VALUE (USD)	REPLACEMENT VALUE (UGX @ 3,700)

1. Sovereign 5-Locale Voice & Call Receptionist	Pipecat WebSocket/LiveKit PBX bridge, `SunflowerASR-51`, `Orpheus-3B` TTS, 5-locale (`en`, `lg`, `sw`, `nyn`, `ach`) digit/USSD verbalisation, EN 301 549 RTT controls, JIT voice consent	**420,000**	**1.55 bn**
2. Statutory RAG, 28 Tax Calculators & MCP	Hybrid Qdrant (`bge-m3` + `mxbai-rerank`), `Sunflower-Qwen3.5-9B`, 169 gazetted acts/practice notes, zero-trust MCP orchestration, and deterministic `Decimal` execution of **FY 2026/27 valuation rules**: (a) *EAC CMA Fourth Schedule* **6-method CIF customs valuation hierarchy** (`customs_valuation_hierarchy = 6`) + BoU mid-rate (`bou_mid_rate`); (b) compounded EAC CET (`0%/10%/25%/35%`) → Excise → `18%` VAT stack plus non-compounded CIF levies (`6%` Import WHT, `1.5%` Infrastructure Levy, `35%` 5–8 yr / `50%` >8–15 yr Used Vehicle Environmental Levy, `15%` UNBS PVoC surcharge above USD 2,000 FOB); (c) *2026 Amendment Acts* (`UGX 200k` vehicle / `UGX 30k` motorcycle transfer stamp duty, `UGX 500k` motorcycle first-registration excise, `UGX 300m` VAT registration threshold, `UGX 2m` `5%` e-receipt VAT refund); and (d) *s.12.2.3 Benefits-in-Kind (BIK)* vehicle valuation `(20% × A × B/365) − D` with **35% p.a. reducing-balance depreciation** on market value `A` and **15%** housing cap	**380,000**	**1.41 bn**
3. 4-Tier Portal Co-Pilot & Vision OCR	Confirmation cards (`ADR-0014`), DOM spotlighting, VLM/Tesseract OCR on EFRIS/Installment screens, 8-family URA error engine, 6 MDA connectors (`EFRIS`, `DTS`, `PRN`, `TIN`, `URSB`, `BWIMS`)	**350,000**	**1.30 bn**
4. Officer Workbench, Audit & DevSecOps	Collision-locked ticket desk, RFC 3161 hash-chained audit ledger, UDPA 2019 / ISO 42001 / NIST AI RMF gates, 14-char NIN DLP redaction, Prometheus/OpenSLO observability	**250,000**	**0.92 bn**
Total Replacement Value	**Full-Stack Sovereign GovTech AI Platform (Turnkey Option A priced at USD 1,170,000 / UGX 4.33 bn)**	**1,400,000**	**5.18 bn**

4. Government of Uganda & NITA-U 6-stage ICT valuation and clearance pipeline (FY 2026/27)

Under the **NITA-U Act (Cap. 97, s. 5)**, the **Public Finance Management Act (PFMA, Cap. 171)**, and the **PPDA Act (Cap. 205)**, GoU determines the allowable value and contract price of an MDA IT system through a 6-stage statutory pipeline before funds are released under **Vote 141**:

STAGE & STATUTORY AUTHORITY	VALUATION TEST APPLIED BY GOU / NITA-U	HOW THIS SYSTEM'S VALUATION & PRICE COMPLY
Stage 1: NITA-U IT Project Clearance & Rationalisation Gate (*NITA-U Act Cap. 97 s. 5; NITA-U Strategic Plan FY 2025/26–2029/30*)	**(a) Non-Duplication** (`UGhub`)**:** Deducts value for any module duplicating national shared services (`NIRA`, `URSB` OBRS, `e-GP`, `UESW`). **(b) Sovereign Hosting:** Mandates NITA-U National Cloud or accredited MDA data centre (URA Nakawa). **(c) Security & PDPO Certification:** Requires `ISO/IEC 27001`, `ISO/IEC 27701`, `ISO/IEC 20000`, and `UDPA 2019` compliance.	Zero duplication: integrates with `NIRA`, `URSB`, `EFRIS`, `DTS`, `BWIMS`, and `UESW` via zero-trust MCP connectors (`ADR-0009`), so 100% of the **193 EM** effort is net-new tax/voice AI. On-prem/NITA-U hosting saves **USD 165,755 (UGX 605 m)** over 5 years vs. foreign GPU cloud (`§E4`, 16-month break-even). Pre-verified STRIDE/PDPO/NIN-DLP compliance anchors Subsystem 4's **USD 250,000** valuation.

STAGE & STATUTORY AUTHORITY	VALUATION TEST APPLIED BY GOU / NITA-U	HOW THIS SYSTEM'S VALUATION & PRICE COMPLY
Stage 2: MoFPED Public Investment Management (PIM / PIP) Gate (*PFMA Cap. 171; Project 1904 Vote 141*)	Discounts 5-year cash flows at Uganda's **10% real social discount rate**; requires **BCR ≥ 1.0**, single-system outlay < **1.0%** of the agency Vote, and a competitive cost per citizen served.	At 5-year TCO Present Worth **USD 1.127 m (UGX 4.11 bn)** (`UGX 1,085 m/yr` annual equivalent), **BCR = 1.0** requires converting just **0.079% (2,170)** of dormant taxpayers or deflecting **849 contacts/day**; expected **BCR = 2.47** (§G2) and **4.4×–9.0× Year-1 Net ROI** (`< 2.5 months` payback). Turnkey price is **0.47%** of Vote 141 (`UGX 917.3 bn`); SaaS is **USD 0.11 (UGX 422) / TIN / yr**.
Stage 3: PPDA Engineer's Estimate & `e-GP` Benchmarking (*PPDA Act Cap. 205; World Bank UDAP* `IDA-68980`)	Establishes the Procuring & Disposing Entity (PDE) Reserve Estimate via **(a) Input Method** (`Effort EM × Rate Card`) and **(b) Analogy Method** against comparable FY 2025/26–2026/27 national ICT contracts on `egpuganda.go.ug`.	**Input:** `193 EM × USD 3,500/EM = USD 675,500` local rebuild floor (+ `USD 432,000` for 4 embedded specialists × 24 mos at `USD 4,500/mo`). **Analogy:** Cross-checked against live NITA-U/UDAP `e-GP` contracts (`UG-NITA-U-562756-CS-QCBS` at **USD 500,000**; `UG-NITA-U-495485-GO-DIR` CERT.UG upgrade at **USD 600,000**). The **USD 480,000** Tier-3 base sovereign licence sits inside the **USD 480k–600k** NITA-U national platform band (`71%` of local rebuild cost).
Stage 4: PPDA QCBS (80/20) & BUBU Tax Netback (*PPDA Evaluation Regulations; BUBU Policy;* `FY2026-27.json`)	Evaluates complex ICT tenders via **Quality and Cost-Based Selection (80% Technical / 20% Financial)** with *Buy Uganda, Build Uganda (BUBU)* preference and domestic tax-retention analysis.	Sovereign Option **A4** scores **4.81 / 5.00** technically vs. **1.98 / 5.00** for foreign SaaS **A2** (§E1). Financially, contracting a resident Ugandan vendor retains **18% VAT** and **6% resident WHT** (*ITA s.119*) in Uganda, avoiding the **15% non-resident WHT** (*s.85*), **5% Digital Services Tax** (*s.86A*), and **18% unclaimable reverse-charge VAT** (*VAT Act s.28*) that add a **20%–38% tax gross-up** to foreign SaaS invoices.
Stage 5: 5-Year Lifecycle TCO, Zero Lock-In & 20% AMC Cap (*NITA-U e-Gov Interoperability & OAG Audit Standards*)	Rejects "low upfront, high lock-in" bids by auditing 5-year TCO, IP ownership, exit cost, and capping post-warranty Annual Maintenance Contracts (AMC) at **15%–22% p.a.**	Full source code, model weights, 10,779-chunk Qdrant corpus, and 28 `Decimal` calculators transfer perpetually with **USD 0 exit cost**. Post-handover AMC is capped at **20% p.a.** (`USD 96k/yr` on base licence; `USD 190k/yr` on turnkey), and embedded transfer shifts Y3–Y5 operations to URA FTEs (`USD 77.6k/yr`).
Stage 6: Milestone-Gated IFMS Disbursement & OAG Audit (*PFMA s. 45; PPDA Contract Management*)	Prohibits upfront lump-sum software payment; requires valid `e-GP` vendor registration, URA `TCC`, `EFRIS FDN` e-invoice, and measurable acceptance gates before MoFPED `IFMS` disbursement.	Enforced by the 5 verifiable milestones **M0–M4** (§F3): **30%** on full artefact custody delivery (`M0`), and **70% at risk** across corpus faithfulness ≥ 0.90 (`M1`), p50 < 500 ms + 100% guardrail soak (`M2`), ≥ 3 live URA channels (`M3`), and 60-day unaided URA team operation (`M4`).

PART E

Alternatives, benefit and price

§15.1.6, §15.2.3, §15.2.5

E1

The alternatives, including do-nothing

SWEBOK §15.1.6 requires that the do-nothing alternative be considered "in most, but not all, situations," and §15.2.3 that the best solution must be among the candidates before it can be chosen. Five mutually exclusive alternatives, all on the **same five-year horizon** as §15.2.5 demands.

	ALTERNATIVE	5-YR COST (USD) E	TIME TO NATIONAL SERVICE	DATA SOVEREIGNTY	LANGUAGE & VOICE	IP / EXIT COST
A0	**Do nothing** – contact centre, counters, portal only	~0 incremental	n/a	Retained	English, text, office hours	n/a
A1	**Add human capacity** – scale officers and contact-centre seats	850,000 – 1,400,000	12–18 months to recruit and train	Retained	Limited by who is hired	n/a
A2	**License a foreign SaaS assistant**	700,000 – 1,900,000	4–8 months	**Taxpayer data leaves Uganda**	Vendor's roadmap, not URA's	**None owned; exit cost total**
A3	**Procure a greenfield custom build**	1,500,000 – 3,200,000	24–36 months	Retained if specified	As specified, at full cost	Owned on delivery
A4	**Acquire sovereign licence + embed Cloake Tech team** *(proposed)*	**1,339,800**	**Prototype live today; national scale inside 24 months**	**Retained – self-hosted, no external AI provider**	**en/lg/sw measured at 100% accuracy; voice in both directions**	**Sovereign perpetual rights on signing; exit cost zero**

A0 is not free. Its cost is the status quo: 2.73 m registered non-contributors, 76% of taxpayers unable to establish what they owe, and – increasingly – taxpayers already asking general-purpose foreign AI tools about Ugandan tax and receiving answers that blend Ugandan rules with foreign law. A0 does not avoid AI-mediated tax guidance; it only ensures the guidance is not URA's.

Multiple-attribute evaluation (SWEBOK §15.6.1, additive weighting). Criteria weighted E: 5-year cost 0.20, time to service 0.15, data sovereignty 0.20, reach (language, voice, literacy) 0.20, IP ownership and exit cost 0.15, auditability of guidance 0.10. Scored 0–5.

	COST	TIME	SOVEREIGNTY	REACH	IP / EXIT	AUDIT	WEIGHTED
A0	5.0	0.0	5.0	1.0	2.5	0.0	**2.58**
A1	2.0	1.5	5.0	1.5	2.5	1.0	**2.40**
A2	2.5	3.5	0.5	3.0	0.0	2.5	**1.98**
A3	1.0	0.5	4.5	4.0	4.5	4.0	**3.05**
A4	**4.1**	**4.9**	**5.0**	**5.0**	**5.0**	**5.0**	**4.81**

A4 leads on every criterion except raw cost, where A0 necessarily wins and A2/A1 are close. Note that A2 – the option that looks cheapest on a monthly invoice – scores lowest overall, because it fails the two criteria a revenue authority cannot trade away: sovereignty over taxpayer data and ownership of the capability.

§15.4.1

E2

Benefit–cost analysis

§15.4.1: "A proposal's financial benefits are divided by its costs. Any proposal with a benefit-cost ratio of less than 1.0 can usually be rejected without further analysis."

Rather than assert a benefit figure, the analysis is inverted: **what is the minimum benefit at which BCR = 1.0?** This requires no estimate from Cloake Tech Limited, only the cost side derived in §D4 and one unit parameter from URA. Annual equivalent cost: **UGX 1,085 m/yr**.

Route (a) — voluntary compliance. Dormant register: 2.73 m registrants **[M, Auditor General FY2024/25]**. Let A = average annual remittance of a newly-remitting small taxpayer P.

A (UGX/yr)	Taxpayers needed for BCR = 1.0	As a share of the 2.73 m dormant register
250,000	4,340	**0.159%**
500,000	2,170	**0.079%**
1,000,000	1,085	**0.040%**

Route (b) — contact deflection. Let C = fully-loaded cost of one human-served contact P.

C (UGX)	Contacts/yr for BCR = 1.0	Per day
2,000	542,500	1,486
3,500	310,000	849
5,000	217,000	595

The system pays for itself if it converts roughly one in every 1,250 dormant registrants per year, or deflects on the order of 850 contacts a day. Against a register of 5.25 m and measured sustained capacity of 214 requests per second, neither threshold is demanding. This is the central economic finding of this document, and it does not depend on any optimistic assumption by Cloake Tech Limited.

Illustrative BCR at a conservative combined case E: 500 contacts/day deflected at C = UGX 3,500 (UGX 639 m/yr) plus 0.079% of the dormant register converted at A = UGX 500,000 (UGX 1,085 m/yr) → benefit UGX 1.724 bn against cost UGX 1,085 m → **BCR ≈ 1.59**.

At the scenario in §3 of the Proposal Summary — a single percentage point of the dormant register, 27,300 taxpayers — the compliance benefit alone is UGX 13.65 bn/yr and **BCR ≈ 12.6**.

§15.4.2

E3

Cost-effectiveness analysis

§15.4.2 gives two forms; the **fixed-cost** form is used: given a fixed annual envelope, which alternative maximises benefit? Envelope: **UGX 1,085 m/yr** (the annual-equivalent cost of A4).

ALTERNATIVE	WHAT UGX 1,085 M/YR BUYS	CONTACTS SERVED / YR	COST PER CONTACT
A1 – add officers	≈ 27.1 fully-loaded officers **[P: UGX 40 m/officer/yr]** at 30 contacts/officer/day × 250 days	203,438	**UGX 5,333**
A4 – this system, low adoption	500 contacts/day	182,500	**UGX 5,945**
A4 – this system, moderate adoption	2,000 contacts/day	730,000	**UGX 1,486**
A4 – this system, high adoption	5,000 contacts/day	1,825,000	**UGX 595**

Crossover: 557 contacts/day. Above roughly 557 contacts a day, the system is more cost-effective per taxpayer served than additional headcount – and unlike headcount its cost-per-contact keeps falling, because guidance is a fixed-cost good (§A3). Below 557/day, headcount wins. This is the honest boundary condition of the whole proposal, and it is an **adoption** threshold, not a technical one.

It also identifies where URA's own effort matters most: promoting the channel is worth more than enlarging it.

§15.5.1

E4

Break-even on infrastructure: on-premises versus cloud

§15.5.1 gives exactly this example – "consider a choice between two cloud service providers… break-even analysis identifies the use level where the costs are the same." Applied to the serving tier, sized at 4 GPUs by §B5 and §F5.

Cash flow streams (§15.1.2), USD E with P unit rates:

	ON-PREMISES	CLOUD (RENTED GPU)
t_0 capital	58,000 (4-GPU A6000-class node)	0
Annual operating	10,545 (power 2,532 · cooling 1,013 · hosting 3,000 · ops 4,000)	55,296 (4 × 720 h × USD 1.60/GPU-h)

Break-even: 58,000 + 10,545·n = 55,296·n → **n = 1.30 years ≈ 16 months.**

Over the five-year horizon: on-premises **USD 110,725** against cloud **USD 276,480** – a saving of **USD 165,755 ≈ UGX 605 m**. Because an always-on taxpayer assistant runs at effectively 100% duty, utilisation is far above the 45% break-even threshold.

On-premises is both cheaper and the only option consistent with the requirement that no taxpayer information reaches an external AI provider. The economics and the policy agree, which is rare and worth stating.

PART F

The price

§15.10.13

F1

Pricing basis

§15.10.13: *"Pricing factors include manufacturing cost, market placement, competition, market condition and product quality."* Each is addressed, and each moves the price:

FACTOR	ASSESSMENT	EFFECT ON PRICE
Manufacturing cost	193 EM converged replacement effort (§D1); USD 675.5 k at the local corporate rate scenario	Sets the commercial benchmark of what the capability costs to rebuild
Market placement	Sovereign public institutions and regulated enterprises across Local (Uganda), East African (EAC), and International tiers	Structured as perpetual sovereign licensing (or Enterprise SaaS) with auditable milestones
Competition	A2 (foreign SaaS) at USD 700 k–1.9 m over five years with zero IP; A3 (greenfield) at USD 1.5–3.2 m and 24–36 months	The price sits clearly below greenfield procurement while preserving data sovereignty
Market condition	Kenya moved first (Apr 2026, text-only WhatsApp); Rwanda and Tanzania have discussed but not deployed	A time-limited window; argues against a price that delays a decision
Product quality	Measured: 100% guardrail defence, 0.93 faithfulness, 100% multilingual FAQ accuracy, 0 cross-tenant violations, 28/28 STRIDE threats mitigated, 12/12 standards mapped	Supports a commercial corporate fee because the assurance work is already done and evidenced

§15.10.2 also notes that *"the target cost can be below the actual estimated cost."* Because Cloake Tech Limited retains core platform IP for regional and international licensing across non-competing jurisdictions, the local sovereign licence fee for URA is set below full standalone replacement cost (§D2).

F2

Recommended local sovereign price (Uganda)

SOVEREIGN ENTERPRISE LICENCE AND MANDATE IP GRANT (LOCAL — UGANDA)

USD 480,000 · UGX 1,752,000,000

Negotiation envelope **USD 395,000 – 550,000** (UGX 1.44 bn – 2.01 bn)

What the fee covers

- Perpetual, irrevocable sovereign licence and mandate IP grant in the delivered system for URA: source code, trained and fine-tuned model weights, fine-tuning datasets, pipeline architecture, the 5,071-passage indexed corpus and its ingestion pipelines, 58 documents, 21 CI/CD pipelines, SBOM and provenance attestations, and the complete evaluation and benchmark artefact set.
- Zero recurring per-seat or royalty fee thereafter within URA's statutory mandate. Exit cost zero.
- Unrestricted right for URA to use, modify, extend, audit, and redeploy within its mandate, while Cloake Tech Limited retains core multi-tenant platform IP for commercial licensing outside URA's mandate.
- Cloake Tech Limited's commitment to a dedicated two-year corporate engineering engagement (priced separately, §F4).

How the figure is justified

REFERENCE	VALUE	FEE AS A SHARE
Replacement cost, Local corporate rate	USD 675,500	**71%**
Replacement cost, East African vendor rate	USD 1,351,000	**36%**
Replacement cost, International integrator rate	USD 2,895,000	**16.6%**
A3 – greenfield procurement, low end	USD 1,500,000	**32%**
Five-year TCO of the whole programme	USD 1,339,800	**36%**
Implied unit rate against 193 converged EM	USD 2,487 / EM	–

What the fee excludes – infrastructure (§F5, URA-provided), the two-year corporate engagement retainer (§F4), applicable taxes (18% VAT / WHT), and the scope extensions of §F6–§F7.

F3

Payment schedule tied to measurable gates

Milestone payments, each released against a gate URA can verify independently. This structure also discharges SWEBOK §15.2.7 – monitoring the performance of the selected alternative – by making the estimates in this document falsifiable at each stage.

	MILESTONE	GATE URA VERIFIES	SHARE	USD
M0	Signing and sovereign licence grant	Agreement executed; full artefact set (code, weights, corpus, docs, pipelines) delivered into URA custody and independently buildable	30%	144,000
M1	URA-corpus optimisation and SME validation	Faithfulness ≥ 0.90 on a URA-authored golden set; URA subject-matter experts sign off on a sampled answer set	20%	96,000
M2	Production hardening and scale acceptance	Measured p50 < 500 ms at the sized concurrency; p95 < 6 s at ceiling; 100% guardrail defence retained; **on-device p95 < 180 ms field-measured on mid-range Android** (closes the package-14 gate, §B3)	20%	96,000
M3	Channel integration live	Serving on ≥ 3 channels (portal, URA app, WhatsApp) with the staff workbench in daily use by contact-centre officers	15%	72,000
M4	Handover complete	URA's own team operating, deploying and extending the system unaided for 60 consecutive days	15%	72,000

M0 at 30% reflects that the majority of the engineering is already complete and transfers on signing; the remaining 70% is at risk against gates that only URA can certify.

F4

Two-year embedded corporate engagement — rate card

Per §5.3 of the Proposal Summary, four Cloake Tech Limited specialists embedded within URA's IT and customer-service departments for 24 months, agreed separately from the licence fee. Corporate rate card P:

TIER	USD / PERSON / MONTH	4 PEOPLE × 24 MONTHS	UGX
Associate AI / Systems Engineer	3,200	307,200	1.12 bn
Mid AI / Systems Engineer (recommended)	**4,500**	**432,000**	**1.58 bn**
Senior / Principal AI Architect	6,500	624,000	2.28 bn

The recommended tier is carried in the TCO of §D4. Its deliverable is not only the production system but the **transfer of operating capability**: at the end of 24 months URA has a trained internal team, which is why the post-handover maintenance line in §D4 is staffed by URA from Year 3 rather than by contract.

F5

Infrastructure — sizing and budget

URA-provided per §5.2 of the Proposal Summary. Sized from the measured envelope, not from instinct.

Demand model E, from the register M:

INPUT	VALUE
Registered taxpayers	5,250,000
Year-1 addressable adoption P	10% → 525,000 users
Sessions per user per year P	4
Turns per session P	5
Annual turns	10.5 m
Filing-season peak-day multiplier P	3×
Peak-hour share of peak day P	12%
Peak load	**2.9 turns/sec**
Measured full-stack p50 (EN) M	2.24 s
Required concurrent generations	**≈ 7**

Seven concurrent generations sits deep inside the measured low-latency zone of a **single** GPU (c ≤ 100, p50 415 ms). Capacity is therefore driven by availability and burst, not by demand:

CONFIGURATION	RATIONALE
2 GPUs baseline, active/active	High availability, not throughput. A single GPU meets Year-1 demand; two ensure no single point of failure.
4 GPUs at filing-season peak	Absorbs the 3× seasonal multiplier with headroom, and covers concurrent voice sessions.
6–8 GPUs by Year 3	At 40% adoption, four times Year-1 volume.
Production node: 4× 48 GB GPU, 512 GB RAM, 4 TB NVMe	Measured footprint is 5,905 MiB VRAM with 4 workers – 12% of a 48 GB card, leaving ample room for larger model tiers
Staging node: 2× 48 GB GPU	§5.2 requires an environment mirroring production
Backing services: PostgreSQL 16, Redis 7.4, Qdrant 1.13	Delivered stateless per 12-factor; embedded SQLite / BM25 / in-memory fall-backs already implemented for degraded operation

Budget: USD 102,000 capex + USD 14,000/yr operating (§D4), on-premises per the break-even result of §E4 (or USD 122,400 capex if procured and warrantied turnkey through Cloake Tech Limited).

F6

Scope tiers, and the extension priced separately

Tiered scope (SWEBOK §15.10.14, prioritisation). Derived from the package ranking of §B4, so that a constrained budget buys the highest-value packages first.

TIER	PACKAGES INCLUDED	EFFORT (EM OF THE 178 IN §B3)	SHARE	LOCAL FEE (USD)	WHAT THE CLIENT GETS
1 – Core	1, 2, 4 (calculators), 9, 10, 11, 12, 13, 16, 18, 19	104	58%	**285,000**	Grounded, cited answers in English on the web portal; guardrails; UDPA compliance; escalation with context; staff workbench; analytics
2 – Standard	Tier 1 + 3, 4 (full), 6, 7, 15, 17	157	88%	**395,000**	Adds agentic orchestration, multilingual en/lg/sw, the full voice pipeline both directions, content operations, personal memory
3 – Full (recommended)	All 19 packages	178	100%	**480,000**	Adds vision and document intelligence, the statutory knowledge graph, offline/edge for rural districts

Effort shares are taken over the 178 EM of §B3; package 4 is split evenly between its eight statutory calculators (Tier 1) and its MCP transport, policy and Tool-RAG selection layer (Tier 2). Note that Tier 1 is priced **above** its pro-rata share of the fee – 58% of the effort at 59% of the price – because package 19 (platform, MLOps and assurance, 18 EM) is not divisible: a Tier-1 purchase still needs the whole CI, SBOM, provenance and deployment estate. Tiers 2 and 3 then price close to or below pro-rata, which is where the incremental argument below comes from.

Why Tier 3 is recommended on economic grounds, not enthusiasm. The step from Tier 2 to Tier 3 costs **USD 85,000** and transfers three packages whose replacement cost at the local corporate rate scenario is **21 EM ≈ USD 73,500**, and at regional vendor rates **USD 147,000**. The capability is already built and tested. Declining Tier 3 does not avoid the cost of those packages — it defers it at a higher unit price. This is the incremental comparison SWEBOK §15.3.4 prescribes.

Priced separately — the internal work-product extension. §2 of the Proposal Summary describes turning the same foundation inward: reading long filings, case records and bulk returns, and drafting structured reports for URA officers. Package 8 (vision and document intelligence, KAval +0.61, measured 40 MiB at 4.82 MB/s) is the foundation for it and is already delivered. The extension itself is priced as a modular add-on in §F7.

F7

Cloake Tech Limited — Multi-market commercial schedule (Local, East Africa, International)

As a registered technology vendor, Cloake Tech Limited licenses the platform across three market tiers — **Local (Uganda)**, **East African (EAC)**, and **International** — under both **Sovereign Perpetual Licensing** (public revenue authorities, central banks, ministries) and **Enterprise SaaS / Managed Private Cloud** (commercial banks, telecoms, utilities, insurers, multinationals).

COMMERCIAL LINE ITEM	LOCAL (UGANDA)	EAST AFRICA (EAC)	INTERNATIONAL
Sovereign Licence — Tier 1 (Core)	USD 285,000	USD 480,000	USD 900,000
Sovereign Licence — Tier 2 (Standard)	USD 395,000	USD 680,000	USD 1,350,000
Sovereign Licence — Tier 3 (Full Suite)	**USD 480,000**	**USD 850,000**	**USD 1,750,000**
Turnkey Localization & System Integration	USD 140k – 220k	USD 250k – 420k	USD 550k – 950k
Embedded Engineering Team (4 FTE × 24 mos)	USD 432,000	USD 672,000	USD 1,152,000
Annual Enterprise Support & SLA (20–22% AMC)	USD 96,000 / yr	USD 170,000 / yr	USD 385,000 / yr
5-Year Sovereign Turnkey TCV	**USD 1.532m – 1.612m**	**USD 2.622m – 2.792m**	**USD 5.377m – 5.777m**
Enterprise SaaS — Onboarding & Integration	USD 35k – 75k	USD 75k – 150k	USD 150k – 300k
Enterprise SaaS — Commercial Plan (Text/Web)	USD 66,000 / yr	USD 120,000 / yr	USD 240,000 / yr
Enterprise SaaS — Omnichannel Voice + Text	USD 150,000 / yr	USD 264,000 / yr	USD 540,000 / yr
Add-On: Additional Language Pack (6 EM)	USD 25,000	USD 50,000	USD 95,000
Add-On: Internal Officer Audit Copilot (14 EM)	USD 65,000	USD 125,000	USD 240,000
Add-On: Custom Core-System / ERP Connector	USD 18,000	USD 38,000	USD 75,000
Outright Exclusive IP Buyout (M&A)	USD 850k – 1.20m	USD 1.80m – 2.50m	USD 3.80m – 5.50m

PART G

Risk, sensitivity, and monitoring

§15.2.6

G1

Sensitivity analysis

§15.2.6 requires that where estimate inaccuracy could change the decision, the engineer consider ranges of estimates and perform a sensitivity analysis. The decision variable is the benefit–cost ratio; the test is whether any plausible parameter movement drives it below 1.0.

PARAMETER VARIED	LOW	CENTRAL	HIGH	EFFECT ON BCR
Converged effort	140 EM	193 EM	230 EM	No effect on BCR — effort sets the fee reference, not the cost stream, and the fee is fixed
Rate scenario	Local	EAC Vendor	International	No effect on the local fee, which is fixed at USD 480,000
Five-year TCO	−20% (USD 1.07 m)	USD 1.34 m	+30% (USD 1.74 m)	Annual equivalent UGX 868 m / 1,085 m / 1,411 m; break-even conversion 0.064% / 0.079% / 0.103% of the dormant register — all far below 1%
Adoption	5%	10%	25%	Cost-per-contact UGX 11,890 / 5,945 / 2,378 — **the only parameter that can push A4 below A1**
UGX/USD	3,400	3,650	4,000	Fee UGX 1.63 bn / 1.75 bn / 1.92 bn; no effect on BCR, since both sides move together
Discount rate	6%	10%	14%	Annual-equivalent cost UGX 1,050 m / 1,085 m / 1,120 m; break-even conversion 0.077% / 0.079% / 0.082%
Infrastructure choice	On-prem	—	Cloud	Five-year TCO +USD 166 k → annual equivalent UGX 1,219 m; break-even conversion 0.079% → 0.089%

The decision is insensitive to every parameter except adoption. Under all cost and rate movements tested, the break-even benefit stays inside 0.064%–0.103% of the dormant register — between one in 970 and one in 1,560 registrants per year. Only a sustained failure of adoption below roughly **557 contacts/day** (§E3) reverses the decision.

This locates the risk precisely, and it is not an engineering risk. It is a channel-adoption risk, and it is mitigated by the things §B4 ranks highest: voice, local language, and presence on the channels taxpayers already use.

§15.2.6

G2

Decision-making under risk

Expected-value analysis across three adoption scenarios, probabilities E:

SCENARIO	P	DEFLECTION	CONVERSION OF DORMANT REGISTER	ANNUAL BENEFIT (UGX)
Low	0.25	400/day	0.04%	1.06 bn
Central	0.50	850/day	0.08%	2.18 bn
High	0.25	2,000/day	0.20%	5.29 bn
Expected value				**2.68 bn**

Expected BCR = 2.68 / 1.085 = 2.47. The low scenario returns BCR ≈ 0.98, just below break-even. At the 5th-percentile outcome, the programme is slightly below break-even, and every scenario above it returns a multiple.

§15.2.7

G3

Monitoring the selected alternative

§15.2.7: *"The software engineer needs to 'close the loop' on estimates by comparing them to the actual outcomes. Otherwise, no one will ever know if the estimates were good."* The following are the estimates in this document that URA should hold Cloake Tech Limited to, with the baseline each is measured against. Each is already instrumented — the analytics, tracing and evaluation packages (13) exist to produce exactly these numbers.

ESTIMATE MADE HERE	BASELINE	REVIEW POINT	WHERE IT IS MEASURED
p50 < 500 ms at sized concurrency	415 ms measured at c = 100	M2, then quarterly	OpenTelemetry per-stage spans; Prometheus
Faithfulness ≥ 0.90 on URA corpus	0.93 on the public corpus	M1, then monthly	Evaluation harness, 9 quality gates
Guardrail defence rate 100%	100% across 1,500-probe soak	M2, then per release	Red-team gate in CI
Contacts deflected per day	Not yet measurable	Monthly from go-live	Analytics dashboard; contact-centre volume comparison
Conversion of dormant registrants	Not yet measurable	Annually	URA register data — **only URA can measure this**
Officer time released	Not yet measurable	Quarterly	Ticket volumes and handling times, staff workbench
Five-year TCO USD 1,339,800	—	Annually	URA finance against the §D4 table

The last three are the estimates most likely to be wrong, and the two that matter most for the BCR can only be closed by URA. Cloake Tech Limited recommends they be written into the engagement as joint reporting obligations rather than left to goodwill.

G4

Assumptions and exclusions

Assumptions carried. UGX 3,650/USD; 10% social discount rate; 4-year GPU economic life; 5-year planning horizon; on-premises hosting; Year-1 adoption of 10% of the register; all three corporate labour rate scenarios; the unit parameters *A* (remittance per newly-compliant taxpayer) and *C* (cost per human-served contact), which only URA holds.

Excluded from every figure in this document. Applicable statutory taxes (18% VAT, WHT); URA-side project management and procurement cost; identity-provider licensing and integration into URA's existing IdP; WhatsApp Business API fees and channel commercial terms; network and connectivity provisioning; URA staff time for subject-matter validation; legal review of the licensing and data-sharing agreement; the internal work-product extension (§F6–§F7); and any integration with the taxpayer registry, EFRIS or the portal, which §5.1 of the Proposal Summary places under URA IT supervision and whose cost is scoped separately in §F7.

What would change the price. A material reduction in scope (the tiers of §F6 are the priced options); a requirement for exclusive IP buyout rather than sovereign mandate licensing (§F7); a requirement for Cloake Tech Limited to supply hardware turnkey (+20% procurement and warranty margin on §F5); or extension to languages beyond en/lg/sw, which is priced per language in §F7.

Summary of the offer

Cloake Tech Limited — Local Sovereign Reference Case (Uganda):

LINE	BASIS	USD	UGX
Sovereign enterprise licence and mandate IP grant	§F2, milestoned §F3	**480,000**	**1,752,000,000**
Two-year embedded corporate engagement, 4 specialists	§F4, recommended tier	432,000	1,576,800,000
Infrastructure, URA-provided	§F5	102,000 capex + 14,000/yr	372 m + 51 m/yr
Five-year total cost of ownership	§D4	**1,339,800**	**4.89 bn**
Annual equivalent cost at 10%	§D4	297,220	1,085,000,000

The recommendation. Alternative A4 scores highest on every criterion except raw cost (§E1), returns an expected benefit–cost ratio of 2.47 (§G2), breaks even on converting roughly one in 1,250 dormant registrants per year (§E2), and is the only alternative that keeps taxpayer data in Uganda while leaving URA with sovereign perpetual operating rights and zero exit cost.

Cloake Tech Limited recommends **Tier 3 at USD 480,000** (for Local Uganda sovereign deployment; **USD 850,000** for East African authorities; **USD 1,750,000** for International deployments per §F7), milestoned per §F3, with the two-year corporate engagement agreed alongside it and infrastructure provided per §F5.

Framework references

All framework citations are to *SWEBOK Guide v4.0*, IEEE Computer Society, 2024 (`docs/PricingFramework/swebok-v4.pdf`):

§	TOPIC	USED IN
1.3.2	Economics of Quality of Service Constraints	§B5
7.1.4	Majority of Maintenance Costs	§D4
15.1.2	Cash Flow	§E4
15.1.4	Equivalence	§A4, §D4
15.1.5	Bases for Comparison	§D4
15.1.6	Alternatives, including do-nothing	§E1
15.1.7	Intangible Assets	§C1
15.1.8	Business Model	§A3
15.2.2	Understand the Real Problem	§A2
15.2.3	Identify All Technically Feasible Solutions	§E1
15.2.4	Define the Selection Criteria	§A4
15.2.5	Evaluate Each Alternative	§E1
15.2.6	Ranges, sensitivity, decision under risk	§G1, §G2
15.2.7	Monitor the Selected Alternative	§F3, §G3
15.3.2–3.5	Economic life, planning horizon, replacement, retirement	§A5
15.4.1	Benefit–Cost Analysis	§D5, §E2
15.4.2	Cost-Effectiveness Analysis	§D5, §E3
15.5.1	Break-Even Analysis	§E2, §E4
15.6.1	Compensatory Techniques (additive weighting)	§B4, §E1
15.7.1–7.7	Identifying and Characterizing Intangible Assets (SIPAC)	§B3, §C1, §C2
15.8.2–8.5	Analogy, Decomposition, Parametric, Multiple Estimates	§D1, §D5
15.9.1	Business Case	§A1
15.9.2	Multiple-Currency Analysis	*How to read the figures*

§	TOPIC	USED IN
15.10.2	Cost and Costing, TCO, sunk cost, opportunity cost	§D3, §D4, §D5, §F1
15.10.11	Product Life Cycle (SPLC vs SDLC)	§D4
15.10.13	Price and Pricing	§A1, §D5, §F1, §F7
15.10.14	Prioritization	§F6, §F7

System evidence referenced

ARTEFACT	SUPPLIES
App/backend/app/tax/data/FY2026-27.json	Statutory FY 2026/27 tax rates, EAC CMA 6-method customs CIF valuation hierarchy, used-vehicle environmental levies, 2026 Amendment Acts, and BIK valuation parameters (§D5)
docs/Reports/MASTER_DOCKER_GPU_BENCHMARK_REPORT_2026-08-21.md	Container throughput, multilingual speech latency, document scaling, export latency, tenant isolation
docs/Reports/SINGLE_GPU_CAPACITY_LIMITS_REPORT_2026-08-21.md	The concurrency curve underlying §B5 and §F5
docs/Reports/SECURITY_COMPLIANCE_STRESS_REPORT_2026-08-21.md	Guardrail defence rate, STRIDE coverage, standards verification
docs/Reports/MULTILINGUAL_FAQ_FULL_STACK_ACCURACY_REPORT_2026-08-21.md	en/lg/sw FAQ accuracy and latency
docs/Reports/MCP_CONVERSATIONAL_PAINPOINTS_ACCURACY_REPORT_2026-08-21.md	Calculator arithmetic precision, schema conformance
docs/Reports/EMOTIONAL_INTELLIGENCE_INTENT_STRESS_REPORT_2026-08-21.md	Intent determination and distress-recognition accuracy
docs/Reports/LOAD_STRESS_ENDURANCE_TEST_REPORT_2026-08-21.md	Peak throughput, multi-format ingestion, CI gate status
docs/EVALUATION_REPORT.md	Faithfulness, citation coverage, the 9 quality gates
docs/MODEL_CARD.md	Model inventory and component parameters
governance/ai_risk_manifest.yaml	NIST AI RMF, ISO/IEC 42001, EU AI Act control status
docs/capstone/week10-standards-relationship-map.md	The 10-standard compliance map and gap status
docs/Reports/URA AI Intelligent Assistant - Proposal Summary.pdf	Engagement terms this document prices (§5.1–5.7)